PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: JadualIQ

Version: 1.0

1. Project Overview

- **Problem Statement:** Time conscious individuals with flexible schedules often make scheduling decisions based on routine rather than real time conditions (Thorhauge et al., 2020). This leads to avoidable costs and inefficiency, particularly in Malaysia, where leaving at the wrong time means paying significantly higher fares on ride hailing platforms like Grab, due to peak hours and rain triggered surges, as well as increased travel time for the traffic (Wong, 2022). Currently, there is no solution that enables the users to identify the optimal departure time within their available time window that minimizes both cost down and travel time from traffic congestion.
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence. (Specifically targeting Time conscious individuals with flexible schedules in Malaysia, including part time workers, university students, and gig flex staff).
- **Proposed Solution Summary:** JadualIQ is a web based AI decision copilot. It processes the user's weekly schedule via natural language, suggests the best time slot, and cross references them with weather and transport data. If it detects a surge risk, it proactively recommends cheaper same day timing blocks. Powered by Z.AI's GLM, it delivers a clear cost saving breakdown and a plain language explanation, ensuring users maximize their daily net earnings.

2. Background & Business Objective

- **Background of the Problem:** Time constrained individuals in Malaysia often plan their days around urgency and gut feel rather than informed decisions. Someone with a fixed commitment in the morning and a flexible task in the afternoon, for example, has no easy way to know that shifting that task by two hours could avoid a rain triggered ride hailing

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platform surge or heavy track. The problem is a disconnected planning experience, users can see their calendar but cannot integrate external information like weather or peak pricing into their decisions. This is further complicated by differing financial situations; some users rely on ride hailing platforms while others depend on public transport, meaning the "best time to go" is not one size fits all but highly dependent on each user's preferred and affordable mode of travel.

- **Importance of Solving This Issue:** Time constrained individuals with limited time and varying financial situations are highly sensitive to recurring transport inefficiencies. JadualIQ bridges the gap between passive calendar information and proactive decision making, turning a schedule into an active cost saving tool while reducing the cognitive burden of manual planning.

- **Strategic Fit / Impact:** JadualIQ targets the "AI for Economic Empowerment" by translating data into financial decisions. It delivers measurable economic outcomes: estimated transport cost before vs after, and total RM saved. In this architecture. In this architecture, Z.AI's GLM is mandatory. A standard rule engine cannot parse messy natural language schedules, weigh the trade off of a weather risk against schedule flexibility, and output a human readable recommendation. The GLM acts as the essential reasoning core.

3. Product Purpose

3.1. Main Goal of the System: To help time conscious individuals in Malaysia with flexible schedules, including part time and gig workers reduce transport costs by recommending one practical, cheaper same day timing switch within their existing schedule when a transport surge risk is detected.

3.2. Intended Users (Target Audience):

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- Malaysian part time worker & gig workers
- University students balancing classes with flexible work
- Users with a mix of fixed and flexible weekly commitments.

4. System Functionalities

4.1. Description: JadualIQ is a web based AI assisted system. Users enter their schedule via a conversational chatbot interface. The system structures it into a weekly calendar, checks weather and transport implications, and uses the GLM to recommend a cost saving same day blocks

4.2. Key Functionalities:

- **Schedule Optimizer**
  - User enters schedule through chatbot
  - JadualIQ structures it into a calendar
  - JadualIQ will create the schedule and suggest some alternative that will help the user in their decision making
- **Weather & Transport Cost Alert**
  - Detects when weather may increase transport cost
  - Especially highlights rain related risk that may contribute to higher ride hailing platforms cost
  - Checks whether a cheaper same day change is available
- **Approval Based Conflict Rescheduler**
  - Event blocks can be moved depending on the priority
  - Fixed commitments remain unchanged
  - Users must approve before changes are applied
- **Explanation Panel**
  - Explains why the recommended switch is better
  - Shows original vs suggested timing
  - Includes cost comparison and limitation notes

4.3. AI Model & Prompt Design

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4.3.1. Model Selection: Z.AI GLM is the core reasoning model. It was chosen because schedule data is inherently unstructured. The GLM is required to interpret ambiguous user intent, reason through the feasibility of shifting time block against weather constraints, and generate a trustworthy, plain language explanation to the user.

4.3.2. **Prompting Strategy:** We utilize multi step constrained prompting. Flow: (1) Parse user input into JSON objects, (2) Identify available time slots, (3) Evaluate candidate same day switch feasibility against external data, (4) Generate the final recommendation. This multi step approach isolates the reasoning tasks, preventing the GLM from outputting off topic content and helps separate reasoning tasks from deterministic validation

4.3.3. **Context & Input Handling**:

JadualIQ accepts natural language schedule input through the chatbot.

Planned handling logic:

- short and moderate inputs are parsed directly
- structured schedule objects are used internally after extraction
- only critical missing information triggers a follow up question

Maximum input size:

As the Z. AI's max input is 499 characters, which is around 80-100 words, our chatbot who accepts the input can only accept 80-100 words.

Behavior when limit is exceeded: Answer: Our system is still going to accept the input, however, it will only return an output saying, "I cannot do unrelated job"

4.3.4. **Fallback & Failure Behavior:** If the GLM returns off topic or hallucinated data, our multiple agents independently validate and cross-filter generated outputs. by

checking time formats and numerical consistency. If it fails , it will show and tell the user error, then tell the user to reprompt

5. User Stories & Use Cases

- **User Stories:**
  1. As a part time worker, I want to just type in my schedule and do not need to manually build a structured planner before getting recommendations.
  2. As a user with limited budget, I want JadualIQ to identify a time block that can reduce my transport cost so that I can save money without changing my whole week.
  3. As a user affected by rainy day travel conditions, I want to know when my original timing may become more expensive so that I can decide whether a same day switch is worth it.
  4. As a cautious user, I want to see a clear explanation of the recommendation so that I can trust the system before accepting the change.

- **Use Cases (Main Interactions):**

Use Case 1: Natural Language Schedule Input

- User enters weekly schedule through chatbot
- System parses and structures it into a 7 day calendar
- User reviews displayed commitments

Use Case 2: Same Day Cost Aware Recommendation

- System compares original timing with feasible same day alternatives
- Checks schedule flexibility and cost implications
- Selects one best cheaper switch if available, in terms of cost and time

Use Case 3: Weather Surge Risk Handling

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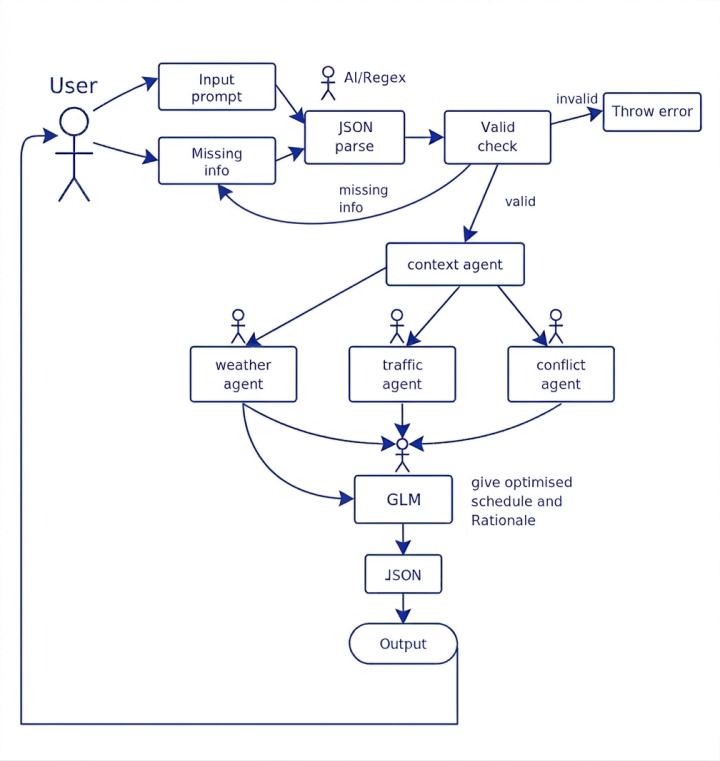
- Rain or adverse weather conditions raise likely transport cost risk
- System checks whether a different same day timing lowers expected cost
- Recommendation is returned as either:
    - Recommended Switch
    - Weather Surge Risk
    - Keep Original
    - Limited Recommendation

Use Case 4: User Approval and Application

- user sees original vs suggested plan
- user accepts the change or keeps the original

- **Use Case Diagram:** (Optional but encouraged for participants to visualize the interaction).



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6. Features Included (Scope Definition)

- Natural language schedule input via chatbot interface
- Structured calendar view with explicit fixed block tagging
- Weather triggered transport cost alerts
- Single best recommendation per switch case
- Recommendation Card containing: Original Timing, Suggested Timing, Estimated ride hailing platforms cost (before/after), Net RM Savings, and Plain Language Explanation.
- "Accept change" and "Keep original" option

7. Features Not Included (Scope Control)

- Mobile app development (limited to responsive web)
- *Production grade gig platform integration (e.g. FoodPanda APIs)*
- *Automated payroll processing*
- *Long term analytics*
- *Production grade public transport integration*
- *Multi user collaboration.*

8. Assumptions & Constraints

- **LLM Cost Constraint:** Cost token around 5000 tokens per week. To keep inference costs manageable, only one active week is processed at a time. Furthermore, prompts are separated into extraction, reasoning and explanation. To reduce repeated raw text submission schedule objects are normalized, only critical follow up questions are asked and repeated internal structures may be reused instead of regenerated.
- **Technical Constraints:**
  - Frontend: JavaScript
  - Backend: Python
  - Database: SQlite
  - Deployment: Web based platform running on a private local environment
  - Reasoning model: Z.AI GLM only

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- public transport remains mocked in the MVP
- weather/maps dependencies may be incomplete during some recommendation attempts.

- **Performance Constraints:** Cross referencing external weather/map logic may introduce minor latency. Quality of the recommendation heavily depends on the detail of schedule provided by the user and only one week is active to keep scope manageable.

- **User Input:** The system assumes the user provides enough basic detail. Ambiguous inputs will force the system to return "I can only help you schedule your activities".

9. Risks & Questions Throughout Development

Risks:

1. **GLM hallucinating numbers or schedule references**
   Mitigation: validator layer checks timing, schedule references, and numerical consistency before user facing output.
2. **Too much manual user input**
   Mitigation: The system ignores every character after the 499th character and generates out of context content.
3. **Weather/maps API failure or incomplete data**
   Mitigation: graceful fallback to Limited Recommendation.
4. **Latency from too many updates**
   Mitigation: strict same day scope and one active week only.
5. **Weak product story if target user becomes too broad**
   Mitigation: keep user groups centered on time constrained Malaysian individuals with flexible schedules, especially part time and gig workers.

Open Questions:

- What minimum input quality is needed before recommendation should proceed?

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A recommendation should only proceed when the system has enough information to compare the original schedule against a same day alternative. Minimum required input should include the activity name, day, approximate time, location or travel context, such as the location. If the input is missing critical details such as day or time, JadualIQ should ask a follow up question instead of generating a recommendation. If external weather or map data is incomplete, the system may still proceed, but the result must be labelled as Limited Recommendation.

- What threshold or logic determines when a recommendation is "worth it" beyond being merely cheaper?

  A recommendation is considered "worth it" when it reduces estimated transport cost by at least RM3 or around 10%, but stays within the wanted date, and produces a clear explanation that the user can act on with one click. The user should not need to manually search for a better time; JadualIQ automatically identifies the better time slot and presents only two actions: Accept Change or Keep Original.

References:

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